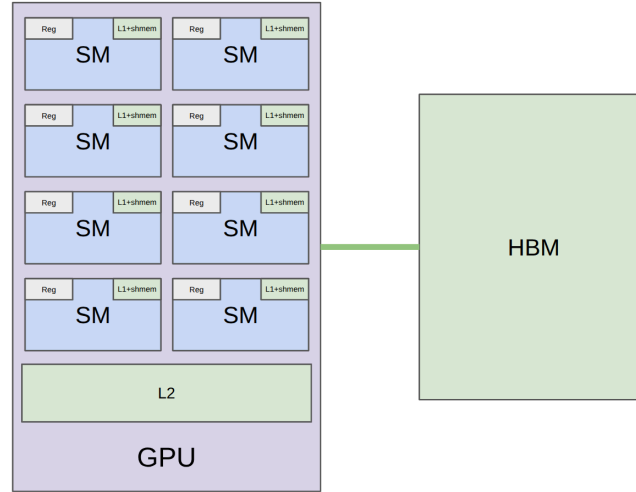


lecture_06.py

```
1 import os
2 import time
3 from typing import Callable
4 import torch
5 from torch.profiler import ProfilerActivity
6 import triton
7 import triton.language as tl
8 from edtrace import text, link, image
9 from lecture_util import get_local_url
10 from gpu_util import cuda_if_available
11
12
13 def main():
14     Last lecture: high-level overview of GPUs and performance
15     This lecture: benchmarking/profiling + writing kernels
16
17     review_of_gpus()
18     benchmarking_and_profiling()      # Where are the bottlenecks?
19     naive_vs_builtin_vs_compiled_gelu() # Apply it to the GeLU example
20
21     # Write Triton kernels
22     triton_introduction()
23     triton_gelu_example()      # Elementwise operation
24     triton_softmax_example()  # Reduction (row fits in a block)
25     triton_row_sum_example()   # Reduction (row doesn't fit in block)
26     triton_matmul_relu_example() # Tiling: use shared memory
27
28     Summary:
29     • Know the programming model (PyTorch, Triton, PTX) to give you correctness
30     • Understand the hardware (SMs, warps, occupancy, bank conflicts, etc.) to optimize performance
31     • Benchmark to understand scaling
32     • Profile to see what's being executed for how long
33     • Triton: think in terms of thread blocks (read to shared memory, do stuff (fusion), write back HBM)
34     • Examples: GeLU (elementwise), softmax (row-wise), row sum (baby tiling), matmul (tiling)
35
36     Next time: more than one GPU!
37
38
39 def review_of_gpus():
```

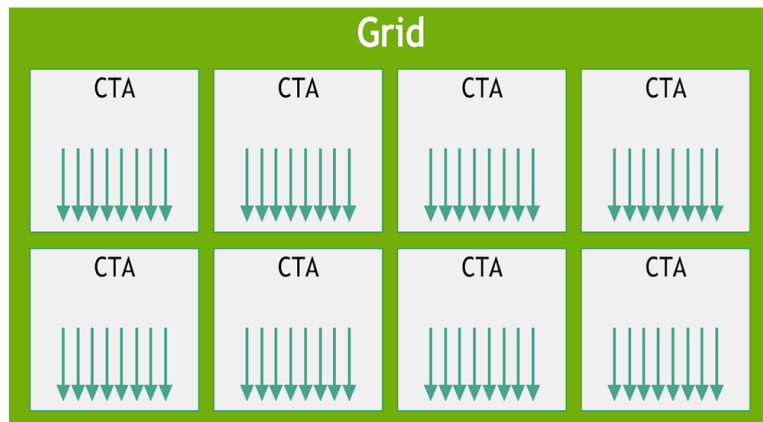
Hardware



Accelerator	A100	H100	B200
# SMs	108	132	148
Register size (per SM)	256 KB	256 KB	256 KB
L1 cache + shared memory (per SM)	192 KB	256 KB	256 KB
L2 cache size	40 MB	50 MB	96-126 MB
HBM size	80 GB	80 GB	192 GB
Register bandwidth	~116 TB/s	~401 TB/s	~447 TB/s
L1 cache + shared memory bandwidth	~19 TB/s	~33 TB/s	~19 TB/s
L2 cache bandwidth	~5-8 TB/s	~12 TB/s	~9 TB/s
HBM bandwidth	2 TB/s	3.35 TB/s	8 TB/s

(B200s also have tensor memory (TMEM) for tensor cores (between registers and shared memory) that are invisible to programmer.)

Programming model



- *Thread*: executes code on a small part of the data
- *Thread block* or concurrent thread array (CTA): a group of threads
- *Grid*: collection of thread blocks

(H100s and B200s also have thread block clusters that enable distributed shared memory.)

Why thread blocks?

For elementwise operations (e.g., GeLU), threads are most natural: each thread processes one element.

- $f(i)$ for $i = 0, \dots, N-1$

However, for non-elementwise operations like softmax or matrix multiplication, threads need to communicate.

Reading/writing from HBM is slow, so use shared memory (local to SM).

Thread block: a collection of threads that access the same shared memory.

Consequently, a thread block is scheduled on one SM.

In Triton, think natively in terms of thread blocks (later).

Interaction between programming model and hardware

Programming model provides an abstraction of the hardware.

In principle, don't need to think about anything else (for correctness).

In practice, performance is very sensitive to the hardware, so need to understand it to obtain high performance.

Let's go over some considerations.

Warps:

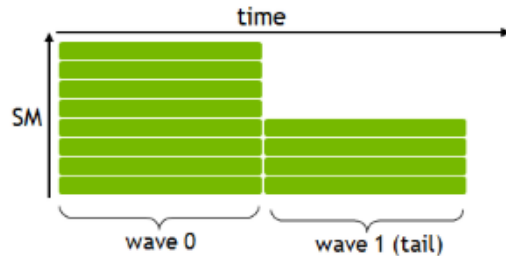
- Within a thread block, threads are grouped into warps (32 threads per warp).

```

85 | Example: thread block has 64 threads => it has 2 warps.
86 | TTTTTTTTTTTTTTTTTTTTTTTTTT | TTTTTTTTTTTTTTTTTTTTTTTTTT |
87 • All threads within a warp must execute same instructions in lockstep on an SM.
88 • Control divergence: if different threads in a warp need to execute different instructions (if A, else B), must be done sequentially (bad)
89 | AAAAAAAAAA..... |
90 | .....BBBBBBBBBBBBBBBBBBBB |
91 • SM runs multiple warps and switches between them (e.g., when one warp is blocked on HBM reads/writes) with zero cost.
92
93 (Warp) occupancy:
94 • Each thread can use between 0 and 255 registers.
95 • The more registers threads use, the fewer threads can be scheduled on an SM (low occupancy).
96 • Low occupancy isn't necessarily bad if each thread is doing more work.
97 • Example: thread coarsening (each thread processes multiple elements).
98 • Example: thread block has 64 threads, each using 160 registers, SM has 65536 registers
99
100 # What we want to run
101 num_threads_per_block = 128
102 num_registers_per_thread = 160
103
104 # What hardware offers
105 max_registers = 65536 # Registers allowed per SM
106 max_warps = 64 # Concurrent warps allowed per SM
107
108 # What we can run at once
109 assert num_registers_per_thread <= 255
110 num_registers_per_block = num_threads_per_block * num_registers_per_thread
111 num_blocks = max_registers // num_registers_per_block # Limited by registers
112 num_warps = num_blocks * num_threads_per_block / 32
113 occupancy = num_warps / max_warps
114
115 Bank conflicts (shared memory):
116 • Shared memory is divided into 32 banks, each 4 bytes wide.
117 B00 B01 B02 B03 B04 B05 B06 B07 B08 B09 B10 B11 B12 B13 B14 B15 B16 B17 B18 B19 B20 B21 B22 B23 B24 B25 B26 B27 B28 B29 B30 B31
118 .....
119 .....
120 • Each cycle, each bank can only be accessed by one thread (if not the same exact location).
121 • If multiple threads access the same bank, accesses serialized (bank conflict).
122 • Worst case example: matrix where each row spans all banks; 32 threads accessing first column results in 32-way bank conflict!
123 • Unavoidable: when doing matmul A @ B, access rows of A and columns of B
124 • Solution: swizzling rearranges shared memory (e.g., row xor col) to avoid bank conflicts
125
126 Memory coalescing (HBM):
127 • When the 32 threads in a warp access HBM, memory accesses combined into transactions of 128 bytes (cache lines).
128 M00 M01 M02 M03 M04 M05 M06 M07 M08 M09 M10 M11 M12 M13 M14 M15 M16 M17 M18 M19 M20 M21 M22 M23 M24 M25 M26 M27 M28 M29 M30 M31
129 M32 M33 M34 M35 M36 M37 M38 M39 M40 M41 M42 M43 M44 M45 M46 M47 M48 M49 M50 M51 M52 M53 M54 M55 M56 M57 M58 M59 M60 M61 M62 M63
130 • Best case: full coalescing, all threads access the same cache line (32 threads x 4 bytes = 128 bytes).
131

```

132 **Block occupancy:**
133



- Thread blocks scheduled onto SMs in waves.
- B200 has 148 SMs, if we launch 160 thread blocks, first wave has 148 blocks, second wave has 12 blocks.
- Wave quantization problem: last wave has fewer thread blocks, leaving some SMs idle (low block occupancy).
- Solution: make number of thread blocks divide # SMs.

139 Summary:

- Programming model: grid (HBM) -> thread block (shared memory) -> thread (registers)
- Details of hardware (warps, bank conflicts, memory coalescing, occupancy) determine performance

144 `def benchmarking_and_profiling():`

145 Recipe for success:

1. Benchmark and profile your code
2. Make changes
3. Benchmark and profile your code again

150 `benchmarking()` # How long does it take?

151 `profiling()` # Where time is being spent?

153 Benchmark and profile your code!

156 `def benchmarking():`

157 Benchmarking measures the wall-clock time of performing some operation.

158 It only gives you end-to-end time, not where time is spent (profiling).

160 It is still useful for:

- comparing different implementations (which is faster?), and
- understanding how performance scales (e.g., with dimension).

164 You can use [torch.utils.benchmark](#).

165 We will roll our own to make benchmarking more transparent.

166

```

167 # Benchmark matrix multiplication
168 matmul = run_operation2(dim=1024, operation=lambda a, b: a @ b)
169 result = benchmark(matmul)
170
171 # See how timing scales with dimension
172 results = {}
173 for dim in [256, 512, 1024, 2048, 4096, 8192]:
174     results[dim] = benchmark(run_operation2(dim=dim, operation=lambda a, b: a @ b))
175
176 Note: time is roughly constant when dimension is small, then cubic scaling.
177
178
179 def benchmark(run: Callable, num_warmups: int = 1, num_trials: int = 3) -> float:
180     """Benchmark `func` by running it `num_trials`. Return the average time."""
181     # Warmup: first times might be slower due to compilation, etc.
182     # Since we will run the kernel multiple times, the timing that matters is steady state.
183     for _ in range(num_warmups):
184         run()
185     torch.cuda.synchronize() # Wait for CUDA threads to finish (important!)
186
187     # Time it for real now!
188     times: list[float] = []
189     for trial in range(num_trials): # Do it multiple times to capture variance
190         # Use CUDA events for accurate GPU timing (avoid capturing CPU overhead)
191         start_event = torch.cuda.Event(enable_timing=True)
192         end_event = torch.cuda.Event(enable_timing=True)
193
194         start_event.record() # Start timing
195         run() # Actually perform computation
196         end_event.record() # End timing
197
198         torch.cuda.synchronize() # Wait for CUDA threads to finish
199
200         times.append((start_event.elapsed_time(end_event)))
201
202     mean_time = mean(times)
203     return mean_time
204
205
206 def profiling():
207     While benchmarking looks at end-to-end time, profiling looks at where time is spent.
208     Independent of time, profiling also helps you understand what's going under the hood.
209
210     PyTorch has a built-in profiler.
211     In your assignment, you will use nsight to get more details.
212
213
214 add(dim=2048)
215 add_profile = profile(run_operation2(dim=2048, operation=lambda a, b: a + b))

```

215

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self
	void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunction_add, std::array<...	0.00%	0.000us	0.00%	0.000us	0.000us	4.9
	Activity Buffer Request	97.35%	3.301ms	97.35%	3.301ms	3.301ms	0
	cudaLaunchKernel	2.02%	68.587us	2.02%	68.587us	68.587us	0
	cudaDeviceSynchronize	0.63%	21.269us	0.63%	21.269us	10.635us	0
Self CPU time total: 3.391msSelf CUDA time total: 4.960us							

216

217

matmul(dim=2048)

218

matmul_profile = profile(run_operation2(dim=2048, operation=lambda a, b: a @ b))

219

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self CUDA
	cutlass3x_sm100_simt_sgemm_f32_f32_f32_f32_f32_64x64x16_1x1x1_3_nnn_align1_bias_f32_relu	0.00%	0.000us	0.00%	0.000us	0.000us	329.345us
	cuLaunchKernelEx	0.95%	27.515us	99.46%	2.867ms	2.867ms	0.000us
	Activity Buffer Request	98.50%	2.839ms	98.50%	2.839ms	2.839ms	0.000us
	cudaDeviceSynchronize	0.54%	15.626us	0.54%	15.626us	7.813us	0.000us
Self CPU time total: 2.883msSelf CUDA time total: 329.345us							

220

```

221 matmul(dim=128)
222 matmul_profile = profile(run_operation2(dim=128, operation=lambda a, b: a @ b))
223

```

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self CUDA
	cutlass3x_sm100_simt_sgemm_f32_f32_f32_f32_f32_32x32x16_1x1x1_3_nnn_align1_bias_f32_relu	0.00%	0.000us	0.00%	0.000us	0.000us	4.544us
	cuLaunchKernelEx	0.78%	32.090us	99.60%	4.118ms	4.118ms	0.000us
	Activity Buffer Request	98.83%	4.085ms	98.83%	4.085ms	4.085ms	0.000us
	cudaDeviceSynchronize	0.40%	16.357us	0.40%	16.357us	8.178us	0.000us

```

Self CPU time total: 4.134msSelf CUDA time total: 4.544us

```

```

224
225 Observations:
226 • You can see which CUDA kernels are actually being called (the long names).
227 • Different CUDA kernels are invoked depending on the tensor dimensions.
228
229 Name of CUDA kernel tells us something about the implementation.
230 Example: cutlass3x_sm100_simt_sgemm_f32_f32_f32_f32_64x64x16_1x1x1_3_nnn_align1_bi...
231 • cutlass: NVIDIA's CUDA library for linear algebra
232 • sm100: corresponds to the NVIDIA Blackwell architecture (B200)
233 • f32: float32
234 • 64x64x16: tile shape (more on this later)
235
236
237 def profile(run: Callable, num_warmups: int = 1):
238     # Warmup
239     for _ in range(num_warmups):
240         run()
241     torch.cuda.synchronize()
242
243     # Run the code with the profiler
244     with torch.profiler.profile(activities=[ProfilerActivity.CUDA],
245                               experimental_config=torch._C._profiler._ExperimentalConfig(verbose=True)) as prof:
246         run()

```



```

247     torch.cuda.synchronize()
248
249     # Print out table
250     table = prof.key_averages().table(sort_by="cuda_time_total",
251                                     max_name_column_width=100,
252                                     row_limit=10)
253
254     # Append to profiles.txt
255     with open("var/profiles.txt", "a") as f:
256         f.write(f"Profile at {time.ctime()}: \n")
257         f.write(table)
258         f.write("\n\n")
259     return table
260
261
262 def naive_vs_builtin_vs_compiled_gelu():
263     Let's benchmark and profile the GeLU activation function.
264
265     x = torch.tensor([1.])
266
267     # 1. Implementation naively from scratch in PyTorch (non-fused)
268     y1 = naive_gelu(x)
269
270     # 2. Built-in PyTorch implementation (fused)
271     y2 = builtin_gelu(x)
272     check_equal_id(naive_gelu, builtin_gelu) # Check it works
273
274     # 3. Use PyTorch compiler on the naive implementation
275     compiled_gelu = torch.compile(naive_gelu)
276     y3 = compiled_gelu(x)
277     check_equal_id(naive_gelu, compiled_gelu) # Check it works (compilation shouldn't change semantics)
278
279     # Benchmarking
280     naive_time = benchmark(run_operation1(dim=16384, operation=naive_gelu))
281     builtin_time = benchmark(run_operation1(dim=16384, operation=builtin_gelu))
282     compiled_time = benchmark(run_operation1(dim=16384, operation=compiled_gelu))
283     The builtin and compiled versions are significantly faster!
284
285     To understand why, let's look at the profiler to see where time is being spent.
286
287     naive_gelu
288     naive_gelu_profile = profile(run_operation1(dim=16384, operation=naive_gelu))

```

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self
void at::native::vectorized_elementwise_kernel<4, at::native::BinaryFuncor<float, float, float, ...		0.00%	0.000us	0.00%	0.000us	0.000us	
	Activity Buffer Request	15.58%	775.108us	15.58%	775.108us	775.108us	0
	cudaLaunchKernel	26.41%	1.314ms	26.41%	1.314ms	145.976us	0
	cudaDeviceSynchronize	58.01%	2.886ms	58.01%	2.886ms	1.443ms	0
Self CPU time total: 4.975msSelf CUDA time total: 3.385ms							

290

291

builtin_gelu

292

builtin_gelu_profile = profile(run_operation1(dim=16384, operation=builtin_gelu))

293

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self
void at::native::vectorized_elementwise_kernel<4, at::native::GeluCUDAKernelImpl(at::TensorIterat...		0.00%	0.000us	0.00%	0.000us	0.000us	
	Activity Buffer Request	77.34%	1.002ms	77.34%	1.002ms	1.002ms	0
	cudaLaunchKernel	21.88%	283.606us	21.88%	283.606us	283.606us	0
	cudaDeviceSynchronize	0.78%	10.070us	0.78%	10.070us	5.035us	0
Self CPU time total: 1.296msSelf CUDA time total: 305.409us							

294

295

compiled_gelu

296

compiled_gelu_profile = profile(run_operation1(dim=16384, operation=compiled_gelu))

	Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	Self CUDA	Self CUDA %	CUDA total	CUDA time avg	# of Calls
	triton_poi_fused_add_mul_tanh_0	0.00%	0.000us	0.00%	0.000us	0.000us	342.848us	100.00%	342.848us	342.848us	1
	Activity Buffer Request	85.72%	1.111ms	85.72%	1.111ms	1.111ms	0.000us	0.00%	0.000us	0.000us	1
	cuLaunchKernel	13.50%	174.982us	13.50%	174.982us	174.982us	0.000us	0.00%	0.000us	0.000us	1
	cudaDeviceSynchronize	0.78%	10.098us	0.78%	10.098us	5.049us	0.000us	0.00%	0.000us	0.000us	2
Self CPU time total: 1.296msSelf CUDA time total: 342.848us											

298

299

Notes:

300

- Naive implementation: multiple kernels, requires many reads/writes from/to HBM (**no fusion**).
- Builtin and compiled versions: one kernel (**kernel fusion**), one read from HBM, one write to HBM.
- The compiled kernel is a Triton kernel.

301

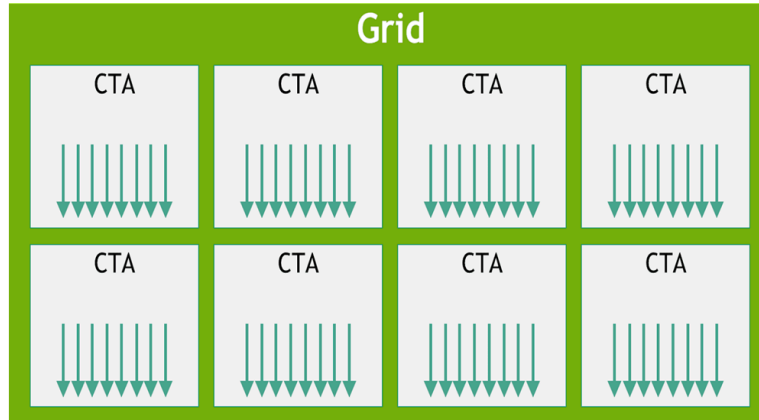
302

303

304

305

def triton_introduction():



307

308

In CUDA (developed by NVIDIA), specify what each thread does.

309

- Pros: fine-grained control

310

- Cons: need to manage more things (e.g., shared memory)

311

312

In Triton (developed by OpenAI), specify what each thread block does.

313

- Generally powerful enough (especially when getting started)

314

- Conceptual framework: load data into shared memory, operate on it, write back to global memory

315

316

317

```
def triton_gelu_example():
```

318

Let's write the Triton kernel for GeLU.

319

320

```
x = torch.randn(8192, device=cuda_if_available())
```

321

```
y = triton_gelu(x)
```

322

323

```
check_equal_1d(triton_gelu, naive_gelu) # Check for correctness
```

324

325

Triton compiles down to PTX (parallel thread execution), an assembly language for GPUs.

326

327

We can see the PTX code generated by Triton.

328

https://github.com/stanford-cs336/lectures/blob/main/var/triton_gelu-ptx.txt

329

330

Observations:

331

- `ld.global.*` and `st.global.*` reads and writes from global memory

332

- `%ctaid.x` is block index, `%tid.x` is thread index

```

333 • %f* are floating point registers, %r* are integer registers
334 • One thread processes 8 elements at the same time (thread coarsening)
335
336
337 def triton_gelu(x: torch.Tensor):
338     # Check input
339     assert x.is_cuda
340     assert x.is_contiguous()
341
342     # Allocate output tensor
343     y = torch.empty_like(x)
344
345     # Determine grid (elements divided into blocks)
346     # | T T T T T T T T | T T T T T T T T | T T T T T T T T | T T T T T T T T |
347     # | Block 0      | Block 1      | Block 2      | Block 3      |
348     num_elements = x.numel()
349     BLOCK_SIZE = 1024 # Number of threads
350     num_blocks = triton.cdiv(num_elements, BLOCK_SIZE)
351
352     # Launch the kernel
353     kernel = triton_gelu_kernel[(num_blocks,)](x, y, num_elements, BLOCK_SIZE=BLOCK_SIZE)
354
355     # Write out PTX (look at this later)
356     output_ptx("triton_gelu", kernel)
357
358     return y
359
360
361 @triton.jit
362 def triton_gelu_kernel(x_ptr, y_ptr, num_elements, BLOCK_SIZE: tl.constexpr):
363     # Input starts at `x_ptr`
364     # Output starts at `y_ptr`
365
366     # | T T T T T T T T | T T T T T T T T | T T T T T T T T | T T T T T T T T |
367     # | Block 0      | Block 1      | Block 2      | Block 3      |
368
369     pid = tl.program_id(axis=0) # Identifies the block
370     start = pid * BLOCK_SIZE     # Starting index of this block
371
372     # Indices where this thread block should operate
373     offsets = start + tl.arange(0, BLOCK_SIZE)
374
375     # Don't read/write past the end of the tensor
376     mask = offsets < num_elements
377
378     # Read
379     x = tl.load(x_ptr + offsets, mask=mask)
380

```

```

381 # Approx gelu is 0.5 * x * (1 + tanh(sqrt(2/pi) * (x + 0.044715 * x^3)))
382 # Compute (t1.tanh doesn't exist, use tanh(a) = (exp(2a) - 1) / (exp(2a) + 1)
383 a = 0.79788456 * (x + 0.044715 * x * x * x)
384 exp = t1.exp(2 * a)
385 tanh = (exp - 1) / (exp + 1)
386 y = 0.5 * x * (1 + tanh)
387
388 # Store
389 t1.store(y_ptr + offsets, y, mask=mask)
390
391
392 def triton_softmax_example():
393     So far, we've looked at elementwise operations in Triton (e.g., GeLU).
394     Now let us look at operations that aggregate over multiple values.
395
396     We will roughly follow the Triton fused softmax tutorial: https://triton-lang.org/main/getting-started/tutorials/02-fused-softmax.html
397
398     Recall the softmax operation is used in attention and generating probabilities.
399     Exponentiate and normalize each row of a matrix:
400     [0 0 0]    =>   [1/3 1/3 1/3]
401     [1 1 -inf]    [1/2 1/2 0 ]
402
403     Let's first start with the naive implementation and keep track of reads/writes.
404     x = torch.tensor([
405         [5., 5, 5],
406         [0, 0, 100],
407     ], device=cuda_if_available())
408     y1 = naive_softmax(x)
409
410     Now let us write the Triton kernel.
411
412     row 0
413     row 1 -- pid=1
414     row 2
415     row 3
416
417     input matrix
418
419     one program instance (pid = row index)
420
421     load row
422     t1.load
423
424     subtract max
425     x - t1.max(x)
426
427     exp + sum
428     t1.exp, t1.sum
429
430     normalize
431     t1.store
432
433     y2 = triton_softmax(x)
434
435     # Check our implementations are correct
436     check_equal_2d(pytorch_softmax, naive_softmax)
437     check_equal_2d(pytorch_softmax, triton_softmax)
438

```

```

419 def naive_softmax(x: torch.Tensor):
420     # M: number of rows, N: number of columns
421     M, N = x.shape
422
423     # Compute the max of each row (MN reads, M writes)
424     x_max = x.max(dim=1)[0]
425
426     # Subtract off the max (MN + M reads, MN writes)
427     x = x - x_max[:, None]
428
429     # Exponentiate (MN reads, MN writes)
430     numerator = torch.exp(x)
431
432     # Compute normalization constant (MN reads, M writes)
433     denominator = numerator.sum(dim=1)
434
435     # Normalize (MN reads, MN writes)
436     y = numerator / denominator[:, None]
437
438     # Total: 5MN + M reads, 3MN + 2M writes
439     # In principle, should have MN reads, MN writes (speedup of 4x!)
440     return y
441
442
443 def triton_softmax(x: torch.Tensor):
444     # Allocate output tensor
445     y = torch.empty_like(x)
446
447     # Determine grid
448     M, N = x.shape # Number of rows x number of columns
449     block_size = triton.next_power_of_2(N) # Each block contains all the columns
450     num_blocks = M # Each block is a row
451
452     # Launch kernel
453     triton_softmax_kernel[(M,)](
454         x_ptr=x, y_ptr=y,
455         x_row_stride=x.stride(0), y_row_stride=y.stride(0),
456         num_cols=N, BLOCK_SIZE=block_size
457     )
458
459     return y
460
461
462 @triton.jit
463 def triton_softmax_kernel(x_ptr, y_ptr, x_row_stride, y_row_stride, num_cols, BLOCK_SIZE: tl.constexpr):
464     assert num_cols <= BLOCK_SIZE
465
466     # Process each row independently

```

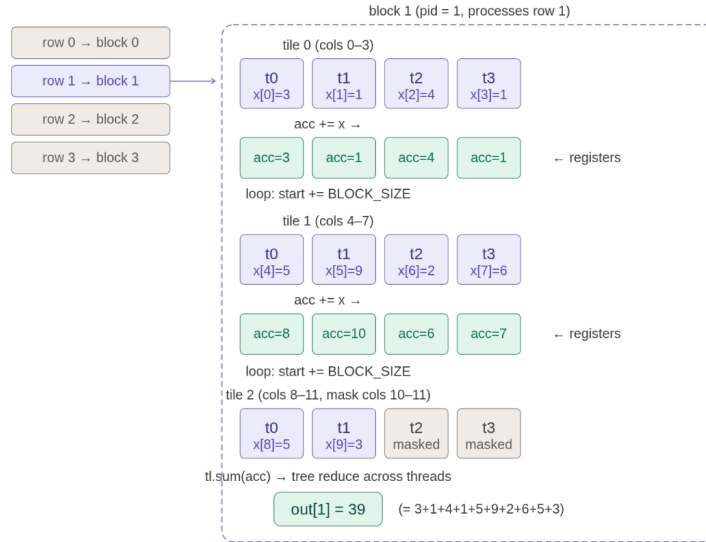
```

467 row_idx = tl.program_id(0)
468 col_offsets = tl.arange(0, BLOCK_SIZE)
469
470 # Read from global memory
471 x_start_ptr = x_ptr + row_idx * x_row_stride
472 x_ptrs = x_start_ptr + col_offsets
473 x_row = tl.load(x_ptrs, mask=col_offsets < num_cols, other=float("-inf"))
474
475 # Compute
476 x_row = x_row - tl.max(x_row, axis=0)
477 numerator = tl.exp(x_row)
478 denominator = tl.sum(numerator, axis=0)
479 y_row = numerator / denominator
480
481 # Write back to global memory
482 y_start_ptr = y_ptr + row_idx * y_row_stride
483 y_ptrs = y_start_ptr + col_offsets
484 tl.store(y_ptrs, y_row, mask=col_offsets < num_cols)
485
486
487 def triton_row_sum_example():
488     In the softmax example, an entire row fits in a block, so the reduction happens within a block (handled by Triton).
489     What if the row doesn't fit in a block?
490     Example: 4096 columns, but block size is 1024...
491
492     Strategy:
493     • Break up row into tiles (4 in the example above)
494     • Each thread iterates over tiles and accumulates a sum
495     • Do final reduction (sum) over accumulators of each thread (shared memory or warp shuffles)
496
497     Consider the simpler example (row sum instead of softmax):
498     x = torch.tensor([[1., 2, 3, 4], [5, 6, 7, 8]], device=cuda_if_available())
499     y1 = builtin_row_sum(x)
500

```


581

input matrix
n_cols = 10



582

```
y2 = triton_row_sum(x)
```

583

584

585

```
def builtin_row_sum(x: torch.Tensor):
```

587

```
    return x.sum(dim=1)
```

588

589

```
def triton_row_sum(x: torch.Tensor, BLOCK_SIZE: int = 1024) -> torch.Tensor:
```

511

```
    M, N = x.shape
```

512

```
    y = torch.empty(M, device=x.device, dtype=x.dtype)
```

513

```
    row_sum_kernel[(M,)](x, y, N, BLOCK_SIZE=BLOCK_SIZE)
```

514

```
    return y
```

515

516

```
@triton.jit
```

517

```
def row_sum_kernel(x_ptr, out_ptr, N, BLOCK_SIZE: tl.constexpr):
```

518

```
    row = tl.program_id(0) # Which row are we processing?
```

519

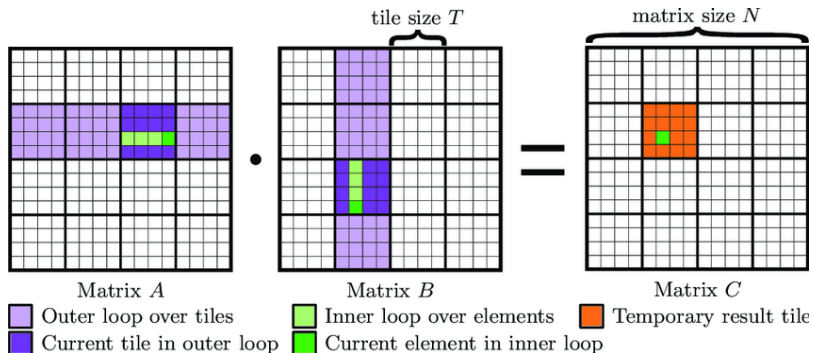
```

520
521 # Accumulator for each thread
522 # One row: T1 T2 T3 T4 | T1 T2 T3 T4 | T1 T2 T3 T4 (N = 12, BLOCK_SIZE = 4)
523 acc = tl.zeros([BLOCK_SIZE], dtype=tl.float32)
524
525 # Loop over tiles
526 for start in range(0, N, BLOCK_SIZE):
527     cols = start + tl.arange(0, BLOCK_SIZE)
528     mask = cols < N
529     x = tl.load(x_ptr + row * N + cols, mask=mask, other=0.0)
530     acc += x
531
532 # Final reduction from BLOCK_SIZE (all threads) to a scalar
533 result = tl.sum(acc, axis=0)
534
535 tl.store(out_ptr + row, result)
536
537
538 def triton_matmul_relu_example():
539     Matrix multiplication is the bread and butter of deep learning.
540     a = torch.randn(1024, 1024, device=cuda_if_available())
541     b = torch.randn(1024, 1024, device=cuda_if_available())
542     c = naive_matmul_relu(a, b)
543
544     How should we build a matmul kernel?
545
546     |           k           n
547     | [ A1 A2 A3 ] [ B1 B2 B3 ] [ C1 C2 C3 ]
548     | m [ A4 A5 A6 ] * k [ B4 B5 B6 ] = [ C4 C5 C6 ]
549     | [ A7 A8 A9 ] [ B7 B8 B9 ] [ C7 C8 C9 ]
550
551     Naive approach:
552     Fix any (m, n).
553     For each k:
554     • Read A[m, k] and B[k, n] from HBM.
555     • Multiply and accumulate.
556     Write result to C[m, n] in HBM.
557
558     Bottleneck: M K N reads, M N writes
559     Arithmetic intensity: O(1)
560
561     Computing C4 and C5 both need A4, A5, A6.
562     Can we read A4, A5, A6 from HBM once to compute both?
563     Answer: yes, using shared memory!
564
565     Idealized approach:
566     • Load all of A and B into shared memory, then compute C.
567     • Now we get M K + K N reads and M N writes.

```

- This yields the idealized $O(N)$ arithmetic intensity from before.
- However, A and B are usually too large to fit in shared memory.

Tiling:



Key idea: divide the matrix C into output tiles (thread blocks).

Fix an output tile in C.

For each pair of (row tile of A, column tile of B):

- Load the corresponding A tile and B tile from HBM into shared memory.
- Perform matrix multiplication on the tiles.
- Accumulate into the partial sum (in shared memory).

Write output tile to HBM.

Arithmetic intensity: $O(\text{tile_size})$.

Bonus:

- Often, you want to apply an elementwise activation function.
- Example: $\text{GeLU}(A @ B)$
- Solution: kernel fusion!

Implementation.

Review: each matrix is linearized in memory

```
x = torch.tensor([[0., 1, 2, 3], [4, 5, 6, 7]])
```

```
stride_row, stride_col = x.stride()
```

```
row = 1
```

```
col = 2
```

```
index = row * stride_row + col * stride_col
```

```
# Compute c = a
```

```
c = triton_matmul_relu(a, b)
```

```

600
601
602 def naive_matmul_relu(x: torch.Tensor, y: torch.Tensor):
603     # Matmul followed by ReLU
604     return torch.nn.functional.relu(x @ y)
605
606
607 def triton_matmul_relu(a: torch.Tensor, b: torch.Tensor):
608     assert a.is_cuda and b.is_cuda
609     assert a.is_contiguous() and b.is_contiguous()
610     assert a.shape[1] == b.shape[0]
611
612     # A is M x K, B is K x N
613     M, K = a.shape
614     K, N = b.shape
615
616     # Allocate output tensor
617     c = torch.empty((M, N), device=a.device)
618
619     # Determine grid
620     BLOCK_M, BLOCK_N, BLOCK_K = 64, 64, 32
621     grid = (triton.cdiv(M, BLOCK_M), triton.cdiv(N, BLOCK_N))
622
623     matmul_relu_kernel[grid](
624         a, b, c,
625         M, N, K,
626         a.stride(0), a.stride(1),
627         b.stride(0), b.stride(1),
628         c.stride(0), c.stride(1),
629         BLOCK_M, BLOCK_N, BLOCK_K,
630     )
631
632     return c
633
634
635 @triton.jit
636 def matmul_relu_kernel(
637     a_ptr, b_ptr, c_ptr,          # Compute c = a
638     M, N, K,                      # a is M x K, b is K x N, c is M x N
639     stride_am, stride_ak,         # How to navigate a
640     stride_bk, stride_bn,         # How to navigate b
641     stride_cm, stride_cn,         # How to navigate c
642     BLOCK_M: tl.constexpr,
643     BLOCK_N: tl.constexpr,
644     BLOCK_K: tl.constexpr,
645 ):
646     # We are working on the (m, n)-th tile
647     pid_m = tl.program_id(0)

```

```

648 pid_n = tl.program_id(1)
649
650 # Indices
651 indices_m = pid_m * BLOCK_M + tl.arange(0, BLOCK_M) # Row indices of a [BLOCK_M]
652 indices_n = pid_n * BLOCK_N + tl.arange(0, BLOCK_N) # Column indices of b [BLOCK_N]
653 indices_k = tl.arange(0, BLOCK_K) # Row indices of a = column indices of b [BLOCK_K]
654
655 # Initial matrix of pointers of a and b
656 a_ptrs = a_ptr + indices_m[:, None] * stride_am + indices_k[None, :] * stride_ak # [BLOCK_M, BLOCK_K]
657 b_ptrs = b_ptr + indices_k[:, None] * stride_bk + indices_n[None, :] * stride_bn # [BLOCK_K, BLOCK_N]
658
659 acc = tl.zeros([BLOCK_M, BLOCK_N], dtype=tl.float32)
660
661 # Move along row tiles of a, column tiles of b
662 for k in range(0, K, BLOCK_K):
663     a = tl.load(a_ptrs, mask=(indices_m[:, None] < M) & (indices_k[None, :] + k < K), other=0.0)
664     b = tl.load(b_ptrs, mask=(indices_k[:, None] + k < K) & (indices_n[None, :] < N), other=0.0)
665     acc += tl.dot(a, b)
666     a_ptrs += BLOCK_K * stride_ak # Advance to the next row tile of a
667     b_ptrs += BLOCK_K * stride_bk # Advance to the next column tile of b
668
669 # Apply activation function (e.g., ReLU)
670 acc = tl.maximum(acc, 0.0)
671
672 # Write output tile
673 c_ptrs = c_ptr + indices_m[:, None] * stride_cm + indices_n[None, :] * stride_cn
674 tl.store(c_ptrs, acc, mask=(indices_m[:, None] < M) & (indices_n[None, :] < N))
675
676
677 #####
678
679 def run_operation1(dim: int, operation: Callable) -> Callable:
680     # Setup: create one random dim x dim matrices
681     x = torch.randn(dim, dim, device=cuda_if_available())
682     # Return a function to perform the operation
683     return lambda : operation(x)
684
685
686 def run_operation2(dim: int, operation: Callable) -> Callable:
687     # Setup: create two random dim x dim matrices
688     x = torch.randn(dim, dim, device=cuda_if_available())
689     y = torch.randn(dim, dim, device=cuda_if_available())
690     # Return a function to perform the operation
691     return lambda : operation(x, y)
692
693
694 def naive_gelu(x: torch.Tensor):
695     # tanh approximation to the gelu activation function

```

```

696 # https://docs.pytorch.org/docs/stable/generated/torch.nn.GELU.html
697 return 0.5 * x * (1 + torch.tanh(0.79788456 * (x + 0.044715 * x * x * x)))
698
699
700 def builtin_gelu(x: torch.Tensor):
701     # PyTorch's built-in GeLU with the tanh approximation
702     return torch.nn.functional.gelu(x, approximate="tanh")
703
704
705 def pytorch_softmax(x: torch.Tensor):
706     return torch.nn.functional.softmax(x, dim=-1)
707
708
709 def check_equal_1d(f1, f2):
710     x = torch.randn(2048, device=cuda_if_available())
711     y1 = f1(x)
712     y2 = f2(x)
713     assert torch.allclose(y1, y2, atol=1e-6)
714
715
716 def check_equal_2d(f1, f2):
717     x = torch.randn(2048, 2048, device=cuda_if_available())
718     y1 = f1(x)
719     y2 = f2(x)
720     assert torch.allclose(y1, y2, atol=1e-6)
721
722
723 def check_equal_2d_2d(f1, f2):
724     x1 = torch.randn(2048, 2048, device=cuda_if_available())
725     x2 = torch.randn(2048, 2048, device=cuda_if_available())
726     y1 = f1(x1, x2)
727     y2 = f2(x1, x2)
728     assert torch.allclose(y1, y2, atol=1e-6)
729
730
731 def mean(xs: list[float]) -> float:
732     return sum(xs) / len(xs)
733
734
735 def output_ptx(name: str, kernel):
736     """Print out the PTX code generated by Triton for the given `kernel`."""
737     ptx_path = f"var/{name}-ptx.txt"
738     with open(ptx_path, "w") as f:
739         ptx = kernel.asm["ptx"]
740         f.write(ptx)
741
742

```